

ENIGMA

Quantum Security Expert // Capstone

Zero to Enterprise-Deliverable in 4 Sessions

~9,200

Lines of Code

353

Tests Passing

20

Python Modules

8/8

Phases Complete

All cryptography built FROM SCRATCH — no library wrappers

Generated: 2026-03-22 18:30

Classification: CONFIDENTIAL // Intellectual Property

TABLE OF CONTENTS

1. Executive Summary
2. Architecture Overview
3. Phase 1: Mathematical Foundations
4. Phase 2: Classical Cryptography
5. Phase 3: Quantum Mechanics
6. Phase 4: Quantum Threat Model
7. Phase 5: Post-Quantum Cryptography
8. Phase 6: PQC Migration Scanner
9. Phase 7: Quantum Key Distribution
10. Phase 8: Portfolio & GTM
11. v2 Enhancements
12. Holy Trinity Audit Results
13. Research Corpus
14. Technical Statistics

1. EXECUTIVE SUMMARY

Enigma is a comprehensive quantum security self-study track that takes a practitioner from zero to enterprise-deliverable expertise across 8 phases. Every cryptographic primitive is implemented from mathematical first principles — no library wrappers, no black boxes. The project demonstrates deep understanding of classical cryptography, quantum computing, post-quantum standards, and security tooling.

The flagship deliverable is **ppqc-scanner** — a multi-language post-quantum cryptography vulnerability scanner with SARIF output for CI/CD integration and certificate chain scanning. It found 156 quantum-vulnerable findings in our own code.

Key achievements:

- ML-KEM (FIPS 203) built from lattice math — NTT, CBD, all 3 parameter sets
- Ed25519 (RFC 8032) from twisted Edwards curve arithmetic
- AES-GCM, RSA-OAEP, ECDSA with RFC 6979, SHA-256/SHA-1 — all from scratch
- Shor's algorithm: real quantum circuit for N=15 on Qiskit
- BB84 QKD: 0% QBER without Eve, 28% with Eve (detected)
- E91: CHSH = 2.96 (quantum violation of Bell inequality)
- CryptoPals Sets 1-4: padding oracle, ECB byte-at-a-time, MT19937 clone
- ppqc-scanner v2: SARIF 2.1.0 + certificate chain scanning
- Holy Trinity audit: 152 findings lifetime, all CRITICALs fixed

2. ARCHITECTURE OVERVIEW

Module Dependency Graph

Module	LOC	Tests	Dependencies
modular_math.py	665	14	None (foundation)
sha256.py	~350	5	None
aes.py	~450	10	None
rsa.py	~400	6	modular_math, sha256
ecc.py	~270	11	modular_math, sha256
ed25519.py	386	24	None (standalone)
cryptopals.py	~400	13	aes
tls_analyzer.py	~350	8	None (network)
quantum_circuits.py	380	13	qiskit
quantum_threats.py	465	11	quantum_circuits
quantum_hardware.py	~300	7	quantum_circuits, qiskit_aer
pqc_lattice.py	~420	10	modular_math, rsa, ecc
ml_kem.py	579	38	None (standalone FIPS 203)
pqc_scanner.py	~580	20	cryptography (certs only)
qkd_protocols.py	371	8	qiskit
portfolio.py	274	6	All phases (stats)

3. Phase 1: Mathematical Foundations

665 LOC | 14 tests

The bedrock. Modular arithmetic, extended GCD, finite fields $GF(p)$ and $GF(2^8)$, AES S-Box construction, Miller-Rabin primality, CRT, Tonelli-Shanks, Pollard's rho.

- modular_math.py: 22 functions covering all crypto math primitives
- $GF(2^8)$ with AES irreducible polynomial $x^8 + x^4 + x^3 + x + 1$
- RSA end-to-end: $p=61$, $q=53$, $e=17$, $\text{encrypt}(65)=2790$

4. Phase 2: Classical Cryptography

~2,400 LOC | 97 tests

Every major classical algorithm from scratch. SHA-256/SHA-1/HMAC, AES (ECB/CBC/CTR/GCM), RSA (PKCS#1/OAEP/signatures), ECC (secp256k1/P-256/ECDH/ECDSA/RFC 6979), Ed25519 (RFC 8032), CryptoPals Sets 1-4, live TLS analyzer.

- AES-GCM: GHASH over $GF(2^{128})$ with authentication tag verification
- RFC 6979: deterministic ECDSA nonces (prevents Sony PS3 class attacks)
- Ed25519: twisted Edwards curve, extended coordinates, HWCD08 doubling
- CryptoPals: padding oracle, byte-at-a-time ECB, MT19937 clone
- TLS: live analysis of google.com (TLSv1.3, AES-256-GCM, PQC detection)

5. Phase 3: Quantum Mechanics for CS

~680 LOC | 20 tests

10 quantum circuits on Qiskit 2.3, teleportation with 0.99+ fidelity, QFT implementation verified unitary ($QFT^4 = I$). Hardware execution module with IBM Eagle/Heron noise model.

- Circuits: superposition, Bell, GHZ, NOT, phase kickback, Deutsch, B-V, Grover
- Teleportation: deferred measurement, partial trace verification
- Hardware: noise-model simulator with ~98% avg success rate across 5 circuits

6. Phase 4: Quantum Threat Model

465 LOC | 11 tests

Shor's factoring (quantum circuit for $N=15$, classical for $N=21/35/77/91$), Grover's search (93-95% on 3 qubits), Mosca's inequality framework, full threat assessment report generator.

- Shor: real quantum circuit with modular exponentiation for $N=15$
- Grover: amplitude amplification with oracle for arbitrary targets
- Threat assessment: HNDL analysis, Mosca's inequality, migration urgency

7. Phase 5: Post-Quantum Cryptography

~1,000 LOC | 48 tests

Regev's LWE encryption from scratch, ML-KEM (FIPS 203) with NTT and all 3 parameter sets, PQC benchmark suite, migration guide generator.

- LWE: noisy linear regression over $\mathbb{Z}_q$, CBD sampling, bit encryption
- ML-KEM: NTT/INTT, basemul with twist factors, K-PKE + FO transform
- ML-KEM-512/768/1024: key serialization per FIPS 203 encoding
- Constant-time decapsulation (`hmac.compare_digest`), domain separation $G(d||k)$

8. Phase 6: PQC Migration Scanner

~780 LOC | 20 tests

THE FLAGSHIP. Multi-language scanner (Python/Java/Go/JS/C/C++/Rust) for quantum-vulnerable cryptography. v2 adds SARIF 2.1.0 and certificate chain scanning.

- Detection: RSA, ECDSA, ECDH, DH, DSA, AES-128, SHA-1, MD5, 3DES, RC4
- Output: text, JSON, HTML (XSS-hardened), SARIF 2.1.0 (CI/CD)
- Certificates: PEM/DER parsing, TLS endpoint scanning, algorithm detection
- Risk scoring: logarithmic scale 0-100, CRITICAL/HIGH/MEDIUM/LOW/INFO
- Self-test: found 156 quantum-vulnerable findings in own Phase 2 code

9. Phase 7: Quantum Key Distribution

371 LOC | 8 tests

BB84 and E91 QKD protocol simulations, QRNG, hybrid QKD+PQC architecture whitepaper.

- BB84: 0% QBER without eavesdropper, 28.1% with Eve (detected via threshold)
- E91: CHSH = 2.96 (exceeds classical bound of 2.0, confirms quantum correlations)
- QRNG: quantum random number generator (documented as simulator-based)
- Hybrid: 3-tier architecture (QKD metro, PQC backbone, classical edge)

10. Phase 8: Portfolio & Go-to-Market

274 LOC | 6 tests

Portfolio generator, competency map, certification assessment, GTM strategy.

- Stats: automated project statistics collection across all phases
- Competency map: 6 domains with skill-level assessment
- Certifications: CompTIA Security+, ISC2 CC, IBM Quantum Dev, CISSP, ISARA
- GTM: consulting (\$150-300/hr), product (pqc-scanner SaaS), employment, hybrid

11. v2 ENHANCEMENTS (Session #4)

Enhancement	LOC	Tests	Status
Ed25519 (RFC 8032)	386	24	COMPLETE + AUDITED
ML-KEM (FIPS 203)	579	38	COMPLETE + AUDITED
SARIF 2.1.0 Output	~150	4	COMPLETE + AUDITED
Certificate Scanning	~150	4	COMPLETE + AUDITED
pytest Migration	14 files	353 total	COMPLETE
Hardware Execution	~300	7	COMPLETE (noise-model)
ML-KEM Serialization	~100	8	COMPLETE

12. HOLY TRINITY AUDIT RESULTS

The Holy Trinity workflow deploys parallel diagnostic agents (devTeam) to audit every module for correctness, security, and code quality. Findings are classified as CRITICAL, HIGH, MEDIUM, or LOW. All CRITICALs and HIGHs are fixed before shipping.

Module	Findings	CRITICALs Fixed	HIGHs Fixed	Status
Session 3 (all phases)	120	10	6	CLEAN
ed25519.py	10	0 (1 FP)	3	CLEAN
ml_kem.py	12	3	3	CLEAN
pqc_scanner.py v2	10	3	3	CLEAN
TOTAL	152	16	15	ALL CLEAN

Key Audit Fixes:

- CRITICAL: hmac.compare_digest for ML-KEM CCA2 decapsulation (timing oracle)
- CRITICAL: CBD bit ordering fixed to LSB-first per FIPS 203 (interoperability)
- CRITICAL: FIPS 203 domain separation $G(d||k)$ across parameter sets
- CRITICAL: SARIF URI relativization for GitHub Code Scanning compatibility
- CRITICAL: XSS prevention in HTML reports (scan_path escaping)
- HIGH: Ed25519 point_double now used in scalar_multiply (was dead code)
- HIGH: SHAKE-128 XOF stream extension (was re-hashing same input)
- HIGH: ML-KEM ciphertext length validation in decrypt

13. RESEARCH CORPUS

Document	Agents	Pages/Size
Phase 1: Math Foundations	6	133 pages
Phase 2: Classical Crypto	7	136 pages
Phase 3: Quantum Mechanics	5	~100 pages
Phase 4: Quantum Threats	5	~50 pages
Phase 5: PQC Standards	6	~50 pages
Ed25519 Implementation Research	1	~30 pages
Scanner v2 Research (SARIF + Certs)	1	77 KB MD

Total: 30+ research agents deployed, 7 master documents produced.

14. TECHNICAL STATISTICS

Metric	Value
Total Source LOC	~9,200
Python Modules	20
Test Cases	353 (352 passing, 1 network skip)
Phases Complete	8/8 + v2 enhancements
Algorithms Implemented	15+ (RSA, AES, ECC, Ed25519, SHA, ML-KEM, LWE, BB84, E91, ...)
Languages Scanned (pqc-scanner)	Python, Java, Go, JS/TS, C/C++, Rust, Config
SARIF Rules	17 quantum-vulnerability rules
Audit Findings (lifetime)	152 (16 CRITICAL, 15 HIGH fixed)
Research Documents	7 (469+ pages)
Sessions	4
Build Tool	Python 3.14, Qiskit 2.3, reportlab

CLASSIFICATION: CONFIDENTIAL // INTELLECTUAL PROPERTY

Owner: Ribbz (Jacob Ribbe) // Enigma Project